
Introduction

The role of the teacher in transdisciplinary education fundamentally differs from the traditional teaching role. The teacher is no longer just a transmitter of knowledge, but a facilitator of learning processes that take place at the intersection of disciplines and society. Transdisciplinary education revolves around jointly exploring and addressing complex societal challenges, and that places different demands on a teacher.

Core Roles of the Teacher

In this type of education, the teacher takes on multiple — often simultaneous — roles:

- **Facilitator of learning** – guiding students in their learning process and encourage self-regulation
- **Mediator between disciplines and society** – connecting academic knowledge with practical knowledge from external partners
- **Coach for personal and professional development** – supporting students in developing skills for collaboration, reflection, and dealing with complexity

On the **Theory** page, you'll learn more about the competencies required of a teacher in transdisciplinary education.

The **Practice** page discusses various forms of transdisciplinary educational practice and the shifting role of the teacher within them.

Theory

Knowledge development on this topic is still emerging, but the following provides a helpful starting point with insights into competencies and an overview of different types of practices.

Competencies You Need

Although there is limited empirical research on what these teaching roles specifically require, recent studies (e.g., Veltman et al., 2022; Oonk et al., 2020; Flavian, 2024) highlight the following competencies:

- **Flexibility and adaptability**

Navigating unpredictable learning processes, shift roles as needed, and tailor your approach to diverse learning needs.

- **Interdisciplinary understanding and content breadth**

They don't need to be an expert in all fields, but you can make connections between disciplines and help students do the same.

- **Mediation and communication skills**

Maintaining relationships with external partners and create a safe space for dialogue between different perspectives.

- **Coaching students**

Supporting students in developing transdisciplinary competencies such as critical thinking, collaboration, and self-reflection.

- **Curiosity and professional development**

Continuously developing oneself and remain open to new forms of education and feedback from practice.

- **Dealing with tensions**

Helping students (and themselves) coping with uncertainty, ambiguity, and tensions that arise when learning together about complex problems.

These competencies take time to develop and highlight the importance of professional space and support within the team and organization.

Not every teacher immediately feels comfortable in this role — it's important to be open about this and to learn together.

Different Practices: Who Provides the Content Expertise?

When designing transdisciplinary education, it's important to consider **who brings in the disciplinary perspectives** — the students

or the teachers.

This directly affects the role as a teacher.

1. Student-intensive education

In broad interdisciplinarity, students bring their own disciplinary knowledge.

The teacher guides the learning process as a *guide on the side* and doesn't need to have all the content expertise themselves. This requires a safe learning environment where students actively share their knowledge and learn from each other.

2. Teacher-intensive education

In narrow interdisciplinarity, teachers provide the various perspectives. They take on a more traditional role (*sage on the stage*) and offer content frameworks or knowledge transfer. Students are relatively more passive in this setup.

Both forms can work, but they require different design choices and teaching skills. Often, a student-intensive approach is preferred because it aligns better with contemporary, active learning and is less vulnerable to teacher absence.

Teaching Formats and Organization: Team Teaching or Not?

To ensure all relevant perspectives are included in the education, **co-teaching with a multidisciplinary team of educators** is a logical choice. However, this is costly, vulnerable, and organizationally complex. Therefore, a common approach is to have **one main facilitator (coach)**, with additional content contributions from others via guest lectures, online resources, or external experts.

Tip:

Design the education collaboratively as a team, but implement it with a compact core group of teachers. Ensure that students, through their own backgrounds and online resources, have access to diverse perspectives. This way, you can do justice to the complexity without running into organizational challenges.

References:

Flavian, H. (2024). Transdisciplinary Teaching: Main Concepts and Perspectives. In Transdisciplinary Teaching in Inclusive Schools: Promoting Transdisciplinary Education for Learners with Special Needs (pp. 1-15). Cham: Springer International Publishing.

Oonk, C., Gulikers, J. T., den Brok, P. J., Wesselink, R., Beers, P. J., & Mulder, M. (2020). Teachers as brokers: adding a university-society perspective to higher education teacher competence profiles. *Higher Education*, 80(4), 701-718/

Veltman, M., Van Keulen, J., & Voogt, J. (2022): Teacher strategies that foster students' boundarycrossing expertise when addressing problems with wicked tendencies, *Journal of Vocational Education & Training*.

Practice

A few examples from practice: **Katusha Sol (University of**

Amsterdam), Jasper ter Schegget (University of Amsterdam), and Philippa Collin (InHolland University of Applied Sciences) share their experiences with the teacher's role in transdisciplinary education.

